

LESSON 8-4 • SESSION 1 • PRACTICE • FORM B

Writing Inequalities

Learning target – I can write an inequality that describes a constraint in a real situation. • 6.EE.B.8

REMEMBER

- **At least** = that number or more ($\geq$). **More than** = greater, not equal ($>$).
- **At most, no more than, maximum** = that number or less ($\leq$). **Less than, fewer than** = $<$.
- Ask: is the boundary number **allowed**? Then choose the symbol.

WORKED EXAMPLE

A van can carry at most 8 riders. Write an inequality for r , the riders.

Step 1 Key words: “at most.” Is 8 allowed? Yes.

Step 2 “At most” means less than or equal to: $r \leq 8$.

Step 3 Test: 8 works, 5 works, 9 does not.

Answer $r \leq 8$

WORDS inequality (*desigualdad*) • at least (*al menos*) • constraint (*restricción*)

PRACTICE

show your work • label units

1. Write an inequality for each phrase. Use x .

a. at least 21 _____

b. at most 6 _____

c. more than 40 _____

d. fewer than 10 _____

e. no more than 50 _____

f. a minimum of 3 _____

2. Write an inequality for each rule. Use the variable given.

Rule	Inequality
A rider must be taller than 48 inches. (h)	
A tent holds at most 6 campers. (c)	
A club needs at least 10 members. (m)	
A ride lasts less than 5 minutes. (t)	

3. **Ticket Sales Goal.** A club meets its goal when it sells at least 200 tickets. Write an inequality for t . Name two solutions.

Hint “At least” lets 200 count.

Answer _____

4. **Phone Storage.** A game needs more than 12 GB of free space. Write an inequality for g . Does exactly 12 GB work?

Hint Is 12 itself allowed?

Show your work.

Answer _____

5. **Trip Chaperones.** Every bus needs at least 5 chaperones. One bus has 4. Write an inequality for c . Does this bus meet the rule?

Show your work.

Answer _____

6. **Uniform Order.** A team gets a bulk price when it orders at least 30 shirts. Write an inequality for s . Name two solutions.

Hint *Decide whether 30 counts before you pick a symbol.*

Answer _____

7. **Read It Back.** An elevator sign reads "Maximum 15 passengers."

Hint *"Maximum" means at most.*

a. Write an inequality for p , the passengers. _____

b. Are 15 passengers allowed? Explain. _____

8. **Ride Rules.** Riders must be at least 44 inches tall and no taller than 78 inches. Let h = height in inches.

a. Write an inequality for the lower limit. _____

b. Write an inequality for the upper limit. _____

c. Name two heights that fit both rules and one that does not. _____

FIND THE ERROR

explain the mistake, then fix it

9. **Jaden's Work.** A class wins a prize when it collects more than 300 cans. Jaden wrote $c \geq 300$ for the winning totals. Explain his mistake. Write the correct inequality.

TEST PRACTICE

choose the best answer

10. "More than 30" is written –

A $x < 30$

B $x \leq 30$

C $x \geq 30$

D $x > 30$

11. "No more than 9" is written –

A $x \leq 9$

B $x < 9$

C $x \geq 9$

D $x > 9$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

t ≥ 200 L	c ≥ 5 M	p < 15 X	s ≥ 30 I	g ≥ 12 Z	g > 12 I	p ≤ 15 T	h ≥ 44 S	c > 5 J	t > 200 Q
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#3	#4	#5	#6	#7a	#8a
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____